

3/15/2023 Wind Tunnel Test Pre-Lab

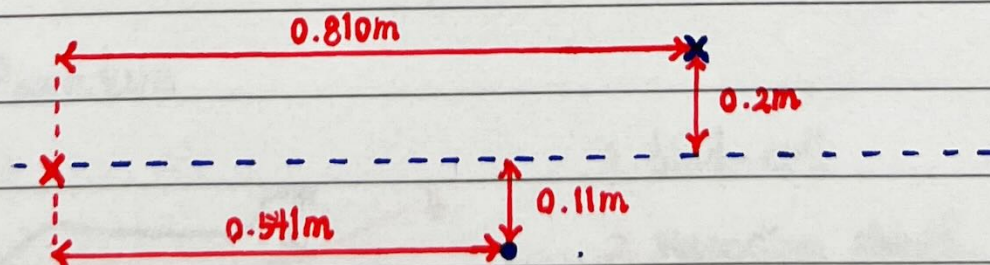
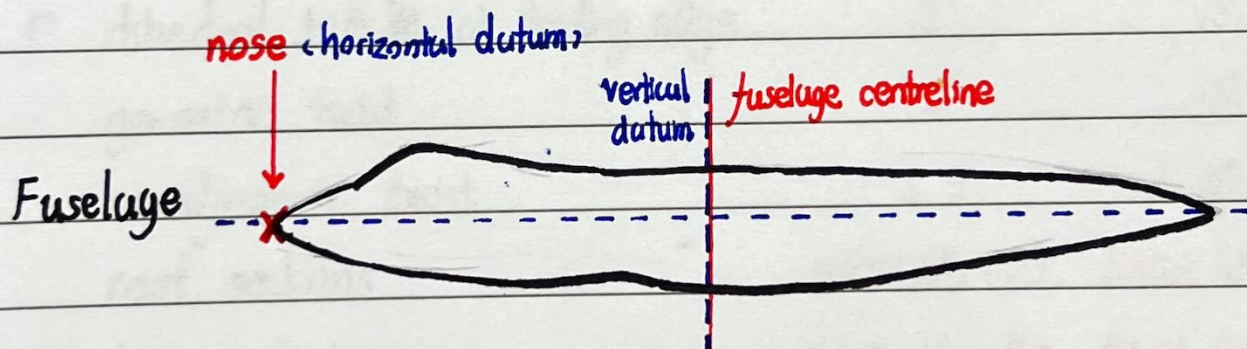
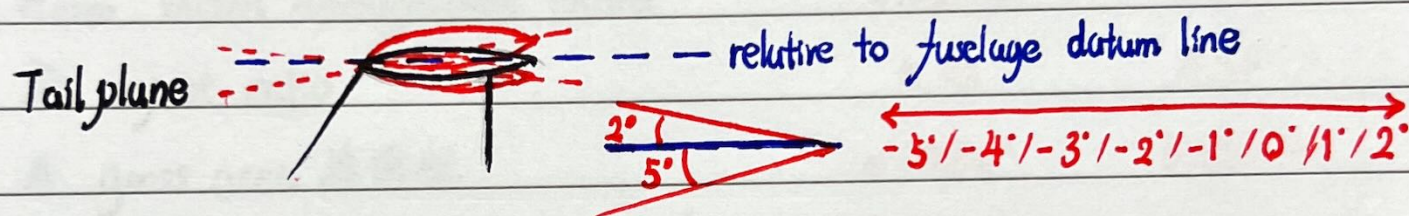
Intro

cryogenic 低温的 wind tunnels ← dynamic similarity

→ Nitrogen gas → $Re_{required}$, $M_{required}$, P_{high} , T_{low}

Model

* see 'The Aircraft' in Flight Simulator Pre-Lab



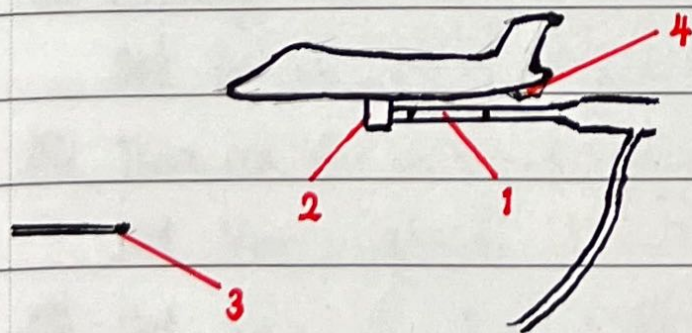
X The quarter-chord point of the tail's mean aerodynamic chord

- The load-cell 称重传感器

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	Wing	Tailplane
b span	0.873 m	0.388 m
C_{root} root chord	0.203 m	0.108 m
C_{cen} chord at aircraft centre-line	0.215 m	0.108 m
C_{tip} tip chord	0.081 m	0.051 m
$\bar{C}_{geo}$ geometric mean chord	0.147 m	0.082 m
$\bar{C}_{aero}$ mean aerodynamic chord	0.158 m	0.085 m
A aspect ratio	5.93	4.74
A gross area 总面积	0.128 m ²	0.032 m ²
Γ dihedral 上反角 at trailing edge	2°	0°
geometric twist	-3°	0°
aerodynamic twist	-3.5°	0°
root sections	NACA 64 ₂ -415	NACA 63,-A012
tip sections	NACA 64 ₂ -412	NACA 63,-A012

Setup & Procedure



- 1 load-cell
- 2 supporting struct
- 3 the Pitot-static tube
- 4 tailplane angle setting pins

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Q. geometric angle of attack = θ string's pitch angle + 0.8°

* only considered F_x +ve backwards (horizontal)

F_2 +ve upwards (vertical)

M_c pitching moment

$$\dot{W}_{\text{motor}} = P_{\text{motor}} \quad 30 \text{ kW}$$

A work $1.6m \times 1.1m$

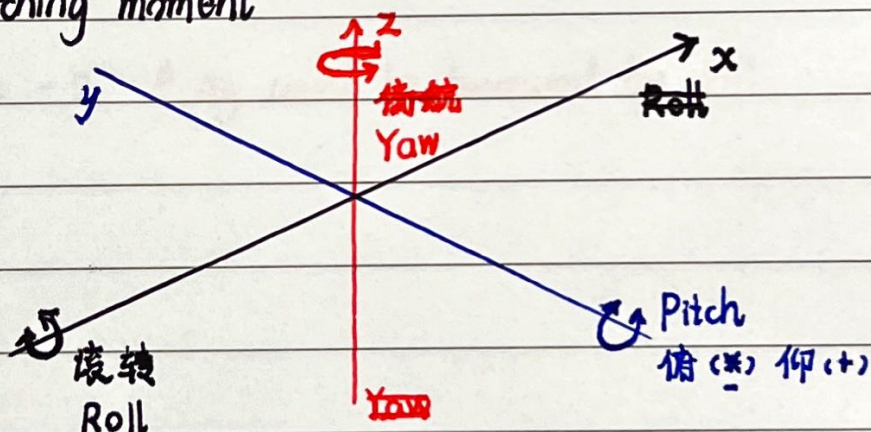
P Patamorphic

v_{max} 40 m/s

$V_{\text{experiment}} = 25 \text{ m/s}$

P.T read from micromanometer

* provide by temperature probe and barometer (p.75)



Procedure

- ① Record T and P * may be varies throughout the lab session
- ② Set $\theta = 0^\circ$
- ③ Ensuring $V_{air} = 0$
set balance reading = 0 (tare 皮重)
- ④ Turn on the wind tunnel
set $V_{air} = 25 \text{ m/s}$ * slowly increase the fan frequency
- ⑤ Set $\alpha_g = -2^\circ$ ($\theta = -2.8^\circ$) * range $-2^\circ \sim 12^\circ$ in steps of 2°

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Record L , D , M_c * 5s average reading

Record standard deviation

⑥ Set for next test

⑦ Reduce V_{air} the adjust tailplane $i_H = +1^\circ$

Repeat procedure

⑧ Repeat procedure $i_H = -2^\circ$ * α_g could be increased by 17°

Avoid model vibrate

⑨ Return $\theta = 0^\circ$

Observe

observe the wool tufts attached to the aircraft

① Onset α_{st} of the wing stall and its spread across the wing

② Location of the wing wake in relation to the tail

③ Stalling of the tail

④ Other...

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* The change of the wool tufts is pre-recorded

IH (deg)	String's pitch angle θ (deg)	Geometric angle of attack α_0 (deg)	Load-cell horizontal axial force		Load-cell vertical axial force		Pitching moment		Temperature		Dynamic Pressure	
			(N)	standard deviation	(N)	standard deviation	(Nm)	standard deviation	($^{\circ}$)	Uncertainty ($\pm$)	(Pa)	Uncertainty ($\pm$)
Remove tailplane	1											
2												

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* Extra test to see what happened when turning / not facing wind

α_H (deg)	String's pitch angle θ (deg)	Geometric angle of attack α_g (deg)	Load-cell horizontal axial force F_x		Load-cell vertical axial force F_z		Pitching moment M_c (My)		Temperature		Dynamic Pressure	
			(N)	standard deviation	(N)	standard deviation	(Nm)	standard deviation	(C°)	Uncertainty ($\pm$)	(Pa)	Uncertainty ($\pm$)